Analytical Cume
(Sept. 29, 2006)

UV-Visible Spectroscopy in Quantitative Analysis

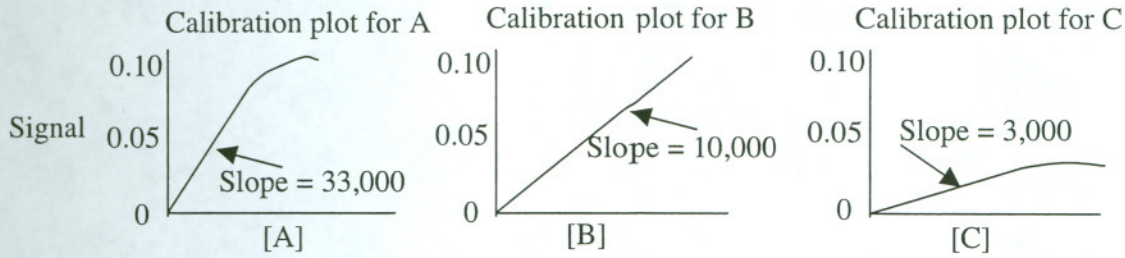
1. Standard Additions (30 points)

The chromium in an aqueous sample was determined by pipetting 10.0 mL of the unknown into each of five 50.0 mL volumetric flasks. Various volumes of a standard containing 12.2 ppm Cr were added to the flasks, and then the solutions were diluted to volume.

A least-squares fit of a plot of absorbance vs. volume of standard (in mL) gave a slope, m, of 0.00881 and a y-intercept, b, of 0.202. Calculate the ppm Cr in the sample.

2. Calibration Curves (40 points total)

An analytical method is used to obtain calibration curves for substance A (Figure A), substance B (Figure B), and substance C (Figure C). Below are some questions about these curves. Be sure to explain why you chose your answers to each of the questions.



(a) (10 points) Order the substances, A, B, and C from highest sensitivity to lowest sensitivity. Explain.
(b) (10 points) Calculate the selectivity coefficient of A relative to B.
(c) (10 points) Which substance provides the largest dynamic range? Why?
(d) (10 points) Three replicates of a blank solution were analyzed, giving the following signals: 0.1, 0.3, and 0.1. What is the approximate detection limit for substance A? (Note: some equations are given below to help, but not all of them are relevant.)

$$\bar{x} = \frac{\sum x_i}{N}$$

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{N-1}}$$

$$S_m = \bar{S}_{blank} + k\, s_{blank}$$

$$C_m = \frac{S_m - \bar{S}_{blank}}{m}$$

$$C_x = \frac{-(V_s)_{s=0}\, C_s}{V_x}$$

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3. UV-Vis in Titrations (30 points total)

Draw the titration curve (relative absorbance vs. volume of titrant) that corresponds to the following scenarios a, b, and c below. Show your work. No credit will be given for a graph without the supporting data and explanation. Assume that the total volume is not significantly changed during the titration (so that you don't have to account for dilution effects.)

Here are the relevant conditions:

- The reaction of interest is: $A + T \rightarrow P$
- Your analyte is "A" and is the only species that absorbs visible radiation in the starting sample solution. "A" does not absorb at 700 nm, but does absorb at 400 nm.
- You titrate your sample solution with the titrant 'T", which absorbs at 700 nm, but does not absorb at 400 nm.
- A product is formed, "P", from the reaction of "A" with "T". "P" has the same absorption characteristics as "A", in that it does not absorb at 700 nm, but does absorb at 400 nm. In addition, it has the same extinction coefficient (molar absorptivity) as "A".
- When one molecule of "A" reacts with one molecule of "T", the reaction is essentially complete and irreversible…a good condition of a titration.

(a) (10 points) Draw the titration curve of absorbance at 400 nm as a function of titrant added. (Also, indicate the endpoint on the titration curve.)

(b) (10 points) Draw the titration curve of absorbance at 700 nm as a function of titrant added. (Also, indicate the endpoint on the titration curve.)

(c) (10 points) What are the advantages of titrations over direct UV-Visible absorbance spectroscopy of a sample of unknown analyte concentration?